

### No. of Functional Features Included in the solution

|                      |                   |
|----------------------|-------------------|
| <b>Date</b>          | 28 September 2026 |
| <b>Team ID</b>       | SWTID-2026-6259   |
| <b>Project Name</b>  | AI StudyBuddy API |
| <b>Maximum Marks</b> | 5 Marks           |

#### Functional Features Overview:

| <b>S.No</b> | <b>Feature Name</b>       | <b>Feature Description</b>                           | <b>Module/Component</b> | <b>Status(Done /In progress/pending)</b> | <b>Marks Contribution</b> |
|-------------|---------------------------|------------------------------------------------------|-------------------------|------------------------------------------|---------------------------|
| 1           | User Registration & login | Allows users to register and securely log in         | Authentication Module   | Done                                     | 1                         |
| 2           | Study Material Management | Allows students to upload and manage study materials | Material Module         | Done                                     | 1                         |
| 3           | AI Summary Generation     | Generates concise summaries from study materials     | Gemini AI Module        | Done                                     | 1                         |
| 4           | AI Flashcards Generation  | Create flashcards for quick revision                 | Gemini AI Module        | Done                                     | 1                         |
| 5           | AI Quiz Generation        | Generates quizzes to test student knowledge          | Gemini AI Module        | Done                                     | 1                         |
| 6           | Personal study plan       | Generates                                            | Study plan              | Done                                     | 1                         |

|   |                           |                                                                     |                 |      |   |
|---|---------------------------|---------------------------------------------------------------------|-----------------|------|---|
|   |                           | study plans based on user requirements and exam dates               | Module          |      |   |
| 7 | Role-based access control | Provides different access privilege for students and administrators | JWT/RBAC Module | Done | 1 |

**Feature Summary:**

| Metric                           | Count/Value |
|----------------------------------|-------------|
| Total featured planned           | 8           |
| Total featured Implemented       | 8           |
| Core/Must-have Features          | 6           |
| Additional/Nice-to-Have features | 2           |
| Features Tested & verified       | 8           |

**Feature Category Breakdown:**

| S.No | Category            | Features in Category | Example features                                               |
|------|---------------------|----------------------|----------------------------------------------------------------|
| 1    | User Interface (UI) | 4                    | Login, Dashboard, study Material,AI features                   |
| 2    | Backend/logic       | 4                    | Authentication, Material Management,AI Processing, Study plans |
| 3    | Database/storage    | 2                    | User Data, study Materials &AI                                 |

|   |                         |   | Resources                                     |
|---|-------------------------|---|-----------------------------------------------|
| 4 | API/Integration         | 2 | REST APIs, Google Gemini API                  |
| 5 | Security/Authentication | 2 | JWT Authentication, role-based access control |